

DD: WO-01-FLOW-SUPPORT — Support Work Order Flow

Developer implementation guide: DD_WO-01-FLOW-SUPPORT-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the operator BPMN flow DD_WO-01-FLOW-SUPPORT-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group III:** Fulfillment & Operations | **Version:** v1.0 | **Module:** Work Order | **Satellite of:** DD_WO-01-Work_Order_Module-v1.0 The Support flow satellite. Defines WorkOrder_Support_WIK.bpmn, the Drools KJAR shape for routing and resolution gating, the operator-config rows seeding the framework's catalogs, the no-site-visit branch for desk-resolved support, the Pattern A two-WO escalation mechanism (finalize original + raise QCS), the area-outage exit, the 3-month warranty + RPT linkage, and the TRC/TRD contractor-billing codes. Built on DD_WO-01-FRAMEWORK-v1.0; reuses parts of DD_WO-01-FLOW-INSTALLATION-v1.0 (equipment binding capture for cases where Support replaces hardware). Covers all Job Types in the SUPPORT kind: GPON support (GP3, GP4, GP2), HFC support (HS1, HS2, HS3, HS4), Wi-Fi extender support (DLC2), TV service call (SRV), proactive support (PS0, PSI), quality control (QCA, QCS), other operational codes (COS, INA, ACD, ROE, LNM, SDI, SDS, SMU), specialized (VIP, ZOI, ZOT, WCS), and the RPT linkage WO when a repeat-within-warranty issue is detected.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/07_work_order/DD_WO-01-FLOW-SUPPORT-v1.0.md
Version	v1.0
DD type	operator BPMN flow
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- POST /api/...
- POST /api/work-orders
- POST /api/work-orders/wo_01HZ...QCS1/finalize-first-confirm
- POST /api/work-orders/wo_01HZ...QCS1/reassign

- POST /api/work-orders/wo_01HZ...SP1/attachments
- POST /api/work-orders/wo_01HZ...SP1/finalize-first-confirm
- POST /api/work-orders/wo_01HZ...SP1/notes
- POST /api/work-orders/wo_01HZ...SP2/finalize-first-confirm

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- contractor-assignment-decision
- r1-vip-always-visit
- r1-vip-internal-first
- r2-no-site-visit-codes
- r2-zuku-office-internal
- r3-default-visit
- r3-qcs-noc-maintenance
- r4-hfc-rf-noc-first
- r5-gpon-by-region
- r6-hfc-by-region
- r7-fallback-default
- resolution-gate
- site-visit-decision
- wo-capture-equipment-bindings-at-finalize
- wo-check-warranty-linkage
- wo-emit-escalation-candidate
- wo-emit-escalation-candidate-if-needed
- wo-emit-state-event
- wo-emit-support-completed
- wo-enforce-finalize-checklist
- wo-mark-escalation-candidate
- wo-tag-warranty-window
- wo-validate-create

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- WorkOrderAssigned
- WorkOrderCancellationRequested
- WorkOrderCreated
- WorkOrderEquipmentBindingsRecorded
- WorkOrderEscalationCandidate
- WorkOrderFinalized
- WorkOrderRPTLinkageRecorded
- WorkOrderSupportCompleted

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 2 | **Group III:** Fulfillment & Operations | **Version:** v1.0 | **Module:** Work Order | **Satellite of:** DD_WO-01-Work_Order_Module-v1.0

The Support flow satellite. Defines `WorkOrder_Support_WIK.bpmn`, the Drools KJAR shape for routing and resolution gating, the operator-config rows seeding the framework's catalogs, the no-site-visit branch for desk-resolved support, the Pattern A two-WO escalation mechanism (finalize original + raise QCS), the area-outage exit, the 3-month warranty + RPT linkage, and the TRC/TRD contractor-billing codes. Built on `DD_WO-01-FRAMEWORK-v1.0`; reuses parts of `DD_WO-01-FLOW-INSTALLATION-v1.0` (equipment binding capture for cases where Support replaces hardware).

Covers all Job Types in the SUPPORT kind: GPON support (GP3, GP4, GP2), HFC support (HS1, HS2, HS3, HS4), Wi-Fi extender support (DLC2), TV service call (SRV), proactive support (PSO, PSI), quality control (QCA, QCS), other operational codes (COS, INA, ACD, ROE, LNM, SDI, SDS, SMU), specialized (VIP, ZOI, ZOT, WCS), and the RPT linkage WO when a repeat-within-warranty issue is detected.

1. What this flow is

A Support WO captures one customer-reported issue from creation through resolution. Caller — the future Ticketing module (creating Support WOs from customer tickets with `originating_context_type=TICKET`), CSR via BO UI (manual creation), or the field contractor (raising a QCS escalation) — supplies the customer, the HomePass, the issue category, the contact info, and any operator metadata. The flow handles assignment, optional dispatch (some Support WOs resolve at the desk without a site visit), on-site work, the resolution gate, equipment binding when hardware is replaced, the QCS escalation pattern when the contractor cannot resolve, and warranty-aware RPT linkage when the issue is a repeat.

Three non-negotiable rules from the workshop:

- 1. Some Support WOs don't need a site visit.** The Job Type catalog marks each code with `requires_site_visit`. Codes like remote-modem-reboot (HS4 Modem Disconnection rare case), Proactive Support (PSO, PSI) when handled remotely, and QCS escalation routed to NOC (NOC resolves remotely or dispatches maintenance themselves) may complete without dispatching a tech. The BPMN's first gateway routes between desk-resolution path and full-on-site-flow path based on the WO's `job_type.requires_site_visit` flag.
- 2. Escalation is two-WO (Pattern A), not single-WO.** Unlike Installation (Pattern B reassignment within same `wo_id`), Support uses a finalize-original-then-spawn-new pattern. The contractor finalizes their support WO with a final reason indicating "could not resolve; escalating" and the framework's flow worker emits `WorkOrderEscalationCandidate`. A new QCS WO is then created — either by the contractor directly (BO UI manual creation) in v1.0, or by the future Ticketing module subscribing to the event and auto-creating. Both paths land at `POST /api/work-orders` with `master_wo_id` set to the original.
- 3. Area-outage is a clean closure with the customer not resolved.** When the field tech determines the issue is an area-wide outage handled separately by NOC's outage-management process, the Support WO closes with `final_reason=AREA_OUTAGE_OPEN`. No QCS is raised — the outage already has its own tracking. The customer remains affected until the outage is fixed (out of this flow's concern).

The Support flow is single-phase — `current_phase` stays NULL throughout. Unlike Shifting, Support does not have sub-phases. Unlike Installation, it does NOT use Pattern B reroute (same `wo_id`, new team) — when work hits a wall, the WO closes and a separate QCS WO opens.

Cross-country variation

This BSS deploys in any Wananchi-Group country. Support flow's cross-country differences are smaller than Installation's (no topology-capture concern — the customer's HomePass already has the topology from their prior install). The variation surface is mostly:

- Job Type catalog rows (each country has its own service codes).
- Final-reason catalog (operator-specific resolution categories).
- Required-before-finalize checklist (operator preferences for evidence).
- Drools rules for assignment + resolution-gate thresholds.

The same per-operator BPMN/Drools/config pattern applies; no flow code changes between operators.

What this flow does NOT do

- **Doesn't create tickets.** The future Ticketing module owns ticket lifecycle (customer reaches out → ticket opened → categorized → routed); Ticketing creates a Support WO when field action is needed. v1.0 ships without Ticketing — Support WOs in v1.0 come from BO UI manual creation by CSRs.

- **Doesn't manage area outages.** Outages have their own NOC process (out-of-scope for v1.0 SOPHIX). The Support flow just provides the AREA_OUTAGE_OPEN exit so contractors can close their visit cleanly while the outage is handled elsewhere.
- **Doesn't compute warranty status.** The wo-check-warranty-linkage worker (§7) queries existing WOs by customer to find a recent install/support within the warranty window; the result drives RPT creation but the warranty calculation itself is pure read against work_order.completed_at + warranty_days. No separate warranty subsystem.
- **Doesn't bill the customer.** Customer billing is unaffected by support visits in v1.0; contractor invoicing (which DOES depend on TRC/TRD/standard codes) is downstream of WorkOrderSupportCompleted event consumed by Finance.
- **Doesn't define the WO module's state machine, REST API, schemas, contractor model, or Kafka primitives** — those live in DD_WO-01-FRAMEWORK.

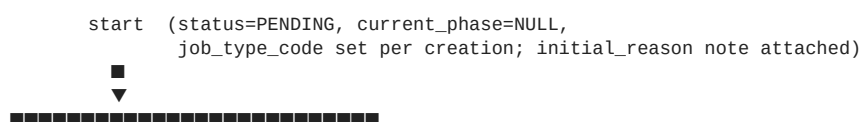
Glossary additions (specific to this flow)

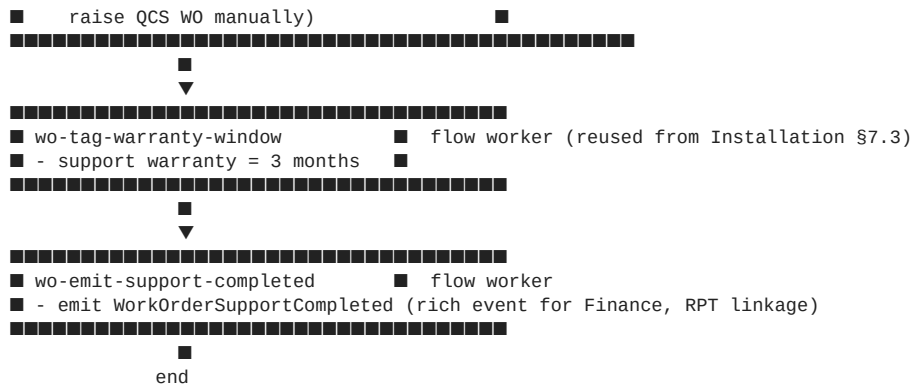
Term	Meaning
No-site-visit branch	BPMN path taken when job_type.requires_site_visit = false. The WO skips contractor assignment + dispatcher claim + on-site work loop and goes straight to a desk-resolution worker that captures resolution notes and finalizes. Used for remote-resolved issues (proactive monitoring, NOC-tier triage, remote modem reboots).
Pattern A escalation	The two-WO support-escalation pattern from the workshop: the original Support WO finalizes with final_reason=COULD_NOT_RESOLVE_ESCALATED (or similar trigger code); the framework's wo-emit-escalation-candidate worker fires; a new QCS WO is created (in v1.0: manually by contractor; future: automatically by Ticketing) with master_wo_id pointing to the original. NOC owns the QCS WO.
QCS WO	Quality Control Service WO. A specific Job Type (QCS) within the Support kind. Created via Pattern A escalation when an original Support WO couldn't resolve. Assigned to the internal NOC Maintenance team (ctr_NOC_MAINT). Runs through the same Support BPMN — same primitives, different team. The QCS code exists specifically to give the NOC team their own trackable WO count (workshop quote: "previously the same WO was passed between contractor and maintenance — making it impossible to separately track the maintenance team's output").
Area-outage exception	When the field tech determines the issue is an area-wide outage, they close the Support WO with final_reason=AREA_OUTAGE_OPEN. No QCS is raised. The flow exits cleanly; the customer remains affected by the outage which is handled by NOC's separate outage-management process.
RPT WO	Repeat Work Order. Created alongside a Support WO (not replacing it) when the new issue is within the warranty window of a prior install (6 months) or prior support visit (3 months). The RPT carries master_wo_id pointing to the prior WO. Both the original new Support WO and the RPT WO progress in parallel; the RPT mostly exists for tracking + contractor billing implications (TRC/TRD codes), not as a separate work item.
TRC / TRD codes	Contractor billing codes used when the contractor returns within the warranty period. TRD = warranty disconnect (contractor takes back equipment installed earlier under warranty); TRC = warranty reconnect (paired with TRD). These codes signal Finance to apply the warranty billing rules (typically no fee or reduced fee depending on whether the original work was at fault). The WO module captures the code; Finance's WorkOrderSupportCompleted consumer interprets it.
Initial Reason	The categorization captured at WO creation that links the WO back to the originating ticket / service request. Workshop: "think of it as the link between a Service Request / Ticket and the WO." Stored as a structured note on the WO (note_kind = initial_reason). Filtered picklist per Job Type.

2. BPMN — WorkOrder_Support_WIK.bpmn

One BPMN per operator (v1.0 ships WIK only). The process key is WorkOrder_Support_WIK; future operators get _WUG, _WTZ, _YASSN. Same workers run; per-operator variation is in operator-config rows (§3) + the operator's Drools KJAR (§4).

2.1 Process shape





+ boundary message catch (interrupting): WorkOrderCancellationRequested
 → wo-mark-cancelled (framework worker) → status=CANCELLED → end-cancelled

The flow has two gateways: site-visit-decision (routes the no-site-visit branch around the contractor assignment + on-site loop) and resolution-gate (decides whether the Drools rules consider the issue resolved or whether the WO closes as an escalation-candidate). The desk-resolution branch is intentionally lean — fewer steps, since there's no field work to coordinate.

2.2 BPMN XML — site-visit-decision gateway

```

<bpmn:businessRuleTask
  id="site-visit-decision"
  name="Drools: requires site visit?"
  camunda:resultVariable="siteVisitDecision"
  camunda:decisionRef="site-visit-decision"
  camunda:decisionRefBinding="latest"
  camunda:mapDecisionResult="singleEntry">
  <bpmn:extensionElements>
    <camunda:inputParameter name="operatorCode">${operatorCode}</camunda:inputParameter>
    <camunda:inputParameter name="jobTypeCode">${jobType}</camunda:inputParameter>
  </bpmn:extensionElements>
</bpmn:businessRuleTask>

<bpmn:exclusiveGateway id="gateway-site-visit">
  <bpmn:outgoing>flow-requires-visit</bpmn:outgoing>
  <bpmn:outgoing>flow-no-site-visit</bpmn:outgoing>
</bpmn:exclusiveGateway>

<bpmn:sequenceFlow id="flow-requires-visit" sourceRef="gateway-site-visit" targetRef="wo-auto-assign-contractor">
  <bpmn:conditionExpression>${siteVisitDecision == 'REQUIRES_VISIT'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>

<bpmn:sequenceFlow id="flow-no-site-visit" sourceRef="gateway-site-visit" targetRef="await-desk-resolution-input">
  <bpmn:conditionExpression>${siteVisitDecision == 'NO_SITE_VISIT'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>

```

The Drools site-visit-decision is trivial — looks up `wo_job_type_catalog.requires_site_visit` for the WO's (operator_code, job_type_code). Returns `REQUIRES_VISIT` or `NO_SITE_VISIT`. Operator-level override is possible by tuning the KJAR (e.g., VIP customers always get a site visit even for codes flagged otherwise).

2.3 BPMN XML — resolution-gate

```

<bpmn:businessRuleTask
  id="resolution-gate"
  name="Drools: resolution outcome"
  camunda:resultVariable="resolutionDecision"
  camunda:decisionRef="resolution-gate"/>

<bpmn:exclusiveGateway id="gateway-resolution">
  <bpmn:outgoing>flow-resolved</bpmn:outgoing>
  <bpmn:outgoing>flow-not-resolved-escalate</bpmn:outgoing>
  <bpmn:outgoing>flow-area-outage</bpmn:outgoing>
</bpmn:exclusiveGateway>

<bpmn:sequenceFlow id="flow-resolved" sourceRef="gateway-resolution" targetRef="wo-capture-equipment-bindings-at-finalize">
  <bpmn:conditionExpression>${resolutionDecision == 'RESOLVED'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>

<bpmn:sequenceFlow id="flow-not-resolved-escalate" sourceRef="gateway-resolution" targetRef="wo-mark-escalation-candidate">
  <bpmn:conditionExpression>${resolutionDecision == 'NOT_RESOLVED_ESCALATE'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>

<bpmn:sequenceFlow id="flow-area-outage" sourceRef="gateway-resolution" targetRef="await-finalize-second-confirm">

```

```
<bpmn:conditionExpression>${resolutionDecision == 'AREA_OUTAGE'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>
```

The Drools resolution-gate reads the WO's final-reason code + the optical/signal readings + the notes. Three outcomes per workshop:

- RESOLVED — issue addressed; equipment bindings captured if hardware was changed; final close.
- NOT_RESOLVED_ESCALATE — issue beyond contractor's scope; escalation candidate flag set; final close with that final_reason; Pattern A QCS spawning happens after.
- AREA_OUTAGE — issue is part of a known area outage; closes cleanly without escalation (no QCS); customer-affecting status handled by NOC's separate outage process.

2.4 Cancellation

Same framework pattern as DD_WO-01-FLOW-INSTALLATION §2.3 and DD_WO-01-FLOW-SHIFTING §2.3 — interrupting boundary message event on WorkOrderCancellationRequested. Valid in any non-terminal status.

2.5 Per-operator BPMN composition

The Support flow varies per operator without code change. v1.0 ships WIK. Future operators can:

- Adjust the no-site-visit eligibility (some operators may always require a tech visit regardless of job type).
- Tune the Drools resolution-gate thresholds (signal-level cutoffs differ by network deployment).
- Add a pre-dispatch ticket-triage user task between WO creation and contractor assignment.
- Change escalation behavior (some operators may use Pattern B reroute for Support, like Installation, instead of Pattern A QCS spawn).

All of these are BPMN edits + config-row additions, not framework or code changes.

3. Per-operator + per-flow configuration rows

This flow seeds the following framework catalogs at v1.0 deploy time. All rows are operator-scoped to WIK.

3.1 wo_flow_config row

operator_code	kind	bpmn_process_key	drools_kjar_ref
WIK	SUPPORT	WorkOrder_Support_WIK	wo-support-wik-v1.0.kjar

3.2 wo_job_type_catalog rows

operator_code	job_type_code	kind	display_name	description	network_type	requires_site_visit	warranty_days
WIK	GP3	SUPPORT	GPON Internet Service Call	LOS, drop cut, internet issues for GPON	GPON	true	90
WIK	GP4	SUPPORT	GPON TV Service Call	TV issues for GPON	GPON	true	90
WIK	GP2	SUPPORT	GPON ADD TV	GPON TV add request	GPON	true	90
WIK	HS1	SUPPORT	HFC Internet Support	Internet issues for HFC	HFC	true	90
WIK	HS2	SUPPORT	HFC TV Support	TV issues for HFC	HFC	true	90
WIK	HS3	SUPPORT	HFC RF / Signal Level Support	RF out of range – maintenance dispatched	HFC	true	90
WIK	HS4	SUPPORT	Modem Disconnection	Rarely used – modem disconnections	HFC	false	90

operator_code	job_type_code	kind	display_name	description	network_type	requires_site_visit	warranty_days
WIK	DLC2	SUPPORT	Wi-Fi Extender Support	Connectivity issue on a Wi-Fi extender	NULL	true	90
WIK	SRV	SUPPORT	TV Service Call	Same purpose as GP4 / HS2 – not commonly used	NULL	true	90
WIK	PS0	SUPPORT	Proactive GPON Service Call	NOC raises proactively from monitoring	GPON	false	0
WIK	PSI	SUPPORT	Proactive HFC Service Call	NOC raises proactively from monitoring	HFC	false	0
WIK	QCA	SUPPORT	Quality Control Audit	Internal QA audit	NULL	false	0
WIK	QCS	SUPPORT	Quality Control Service	Escalation W0 – maintenance team owns this	NULL	true	0
WIK	QOS	SUPPORT	Quality of Service	Customer-facing QoS issue	NULL	true	90
WIK	COS	SUPPORT	Change of Service	Service plan change request	NULL	false	0
WIK	WCS	SUPPORT	Welcome Call Support	Issue reported on the 2-day welcome call	NULL	true	90
WIK	VIP	SUPPORT	VIP Support	MPs, governors, senators, MCAs	NULL	true	90
WIK	ZOI	SUPPORT	Zuku Office Internet	Internet at Zuku office locations	NULL	true	90
WIK	ZOT	SUPPORT	Zuku Office TV	TV at Zuku office locations	NULL	true	90
WIK	RPT	SUPPORT	Repeat Work Order	Repeat-within-warranty linkage W0 (raised alongside another Support W0)	NULL	false	0
WIK	TRC	SUPPORT	Warranty Reconnect	Contractor billing code – paired with TRD	NULL	true	0
WIK	TRD	SUPPORT	Warranty Disconnect	Contractor billing code – paired with TRC	NULL	true	0
WIK	LMN	SUPPORT	Line Maintenance	Maintenance team line-level work	NULL	true	0

3.3 wo_note_kind_registry rows (support-specific kinds)

Inherits framework's generic kinds + reuses installation-flow-registered kinds (findings, solution, optical_readings, hfc_signal_readings, csr_note, dispatcher_note, customer_appointment, final_reason_set). Adds:

operator_code	note_kind	schema_jsonb (excerpt)	append_only
WIK	initial_reason	{"required":["reasonCode","reasonDescription"],"properties":{"reasonCode":{"type":"string"},"reasonDescription":{"type":"string"},"ticketRef":{"type":"string"}}}	true
WIK	warranty_linkage	{"required":["priorWoId","priorJobTypeCode","priorCompletedAt","warrantyDays"],"properties":{"priorWoId":{"type":"string"},"priorJobTypeCode":{"type":"string"},"priorCompletedAt":{"type":"string","format":"date-time"},"warrantyDays":{"type":"integer"}}}	true
WIK	escalation_candidate_reason	{"required":["whyNotResolved","handoffSummary"],"properties":{"whyNotResolved":{"type":"string"},"handoffSummary":{"type":"string"}}}	true
WIK	area_outage_ref	{"required":["outageId"],"properties":{"outageId":{"type":"string"},"outageDescription":{"type":"string"}}}	true
WIK	desk_resolution_summary	{"required":["actionTaken"],"properties":{"actionTaken":{"type":"string"},"customerNotifiedAt":{"type":"string","format":"date-time"}}}	true

3.4 wo_finalization_requirements rows

Two flavors: site-visit codes vs no-site-visit codes. The framework's wo-enforce-finalize-checklist reads the row matching the WO's (operator_code, kind, job_type_code).

operator_code	kind	job_type_code	required_note_kinds	required_attachment_categories	min_attachments_per_category
WIK	SUPPORT	NULL	["initial_reason", "findings", "solution", "final_reason_set"]	[]	NULL
WIK	SUPPORT	GP3	["initial_reason", "findings", "solution", "optical_readings", "final_reason_set"]	["setup_photo", "rf_optical_reading"]	NULL
WIK	SUPPORT	GP4	["initial_reason", "findings", "solution", "optical_readings", "final_reason_set"]	["setup_photo"]	NULL
WIK	SUPPORT	HS1	["initial_reason", "findings", "solution", "hfc_signal_readings", "final_reason_set"]	["setup_photo", "speed_test", "rf_optical_reading"]	NULL
WIK	SUPPORT	HS2	["initial_reason", "findings", "solution", "hfc_signal_readings", "final_reason_set"]	["setup_photo"]	NULL
WIK	SUPPORT	HS3	["initial_reason", "findings", "solution", "hfc_signal_readings", "final_reason_set"]	["rf_optical_reading"]	NULL
WIK	SUPPORT	HS4	["initial_reason", "desk_resolution_summary", "final_reason_set"]	[]	NULL

operator_code	kind	job_type_code	required_note_kinds	required_attachment_categories	min_attachments_per_category
WIK	SUPPORT	PSO	["initial_reason", "desk_resolution_summary", "final_reason_set"]	[]	NULL
WIK	SUPPORT	PSI	["initial_reason", "desk_resolution_summary", "final_reason_set"]	[]	NULL
WIK	SUPPORT	QCA	["initial_reason", "desk_resolution_summary", "final_reason_set"]	[]	NULL
WIK	SUPPORT	QCS	["initial_reason", "findings", "solution", "optical_readings", "hfc_signal_readings", "final_reason_set"]	["rf_optical_reading"]	NULL
WIK	SUPPORT	COS	["initial_reason", "desk_resolution_summary", "final_reason_set"]	[]	NULL
WIK	SUPPORT	RPT	["initial_reason", "warranty_linkage", "final_reason_set"]	[]	NULL

For the QCS code: required notes include both optical_readings AND hfc_signal_readings because the NOC may handle either network type. The Drools rules in §4 deal with which set actually applies per the WO's network_type; this row just lists the kinds that COULD be required. In practice, only one set is checked because only one is captured.

3.5 wo_final_reason_catalog rows

operator_code	kind	job_type_code	final_reason_code	display_name	triggers_escalation_event
WIK	SUPPORT	NULL	RESOLVED_ON_SITE	Resolved on site	false
WIK	SUPPORT	NULL	NO_FAULT_FOUND	No fault found	false
WIK	SUPPORT	NULL	CABLE_REPLACED	Faulty cable replaced	false
WIK	SUPPORT	NULL	MODEM_REPLACED	Faulty modem replaced	false
WIK	SUPPORT	NULL	ONT_REPLACED	Faulty ONT replaced	false
WIK	SUPPORT	NULL	CUSTOMER_NOT_AVAILABLE	Customer not available – reschedule	false
WIK	SUPPORT	NULL	AREA_OUTAGE_OPEN	Area outage – handled separately by NOC	false
WIK	SUPPORT	NULL	COULD_NOT_RESOLVE_ESCALATED	Could not resolve – escalate to NOC	true
WIK	SUPPORT	NULL	RESOLVED_REMOTELY	Resolved remotely (no site visit needed)	false
WIK	SUPPORT	QCS	MAINTENANCE_RESOLVED	Maintenance team resolved upstream issue	false
WIK	SUPPORT	QCS	DISPATCHED_TO_CONSTRUCTION	Issue requires construction work – handed off	false
WIK	SUPPORT	RPT	RPT_LINKAGE_RECORDED	Repeat-within-warranty linkage recorded	false

COULD_NOT_RESOLVE_ESCALATED is the one final reason whose `triggers_escalation_event` is true. When the original Support WO finalizes with this code, the framework's `wo-emit-escalation-candidate-if-needed` worker fires `WorkOrderEscalationCandidate` (event from framework §7). Pattern A handling proceeds (§5).

3.6 Operator variation surface

Same six axes as documented in DD_WO-01-FLOW-INSTALLATION §3.6. The Support flow does not introduce new variation axes.

4. Drools KJAR — `wo-support-wik-v1.0.kjar`

Three named decisions in this KJAR:

1. `contractor-assignment-decision` — picks contractor + team for site-visit Support WOs. Same shape as Installation flow's; routing rules tuned for Support call-out patterns.
2. `site-visit-decision` — reads `wo_job_type_catalog.requires_site_visit` for the WO's (`operator_code`, `job_type_code`). Returns `REQUIRES_VISIT` or `NO_SITE_VISIT`. Operator can override per VIP rules.
3. `resolution-gate` — at first-confirm, decides whether the WO closes as resolved, as escalation candidate, or as area-outage.

4.1 contractor-assignment-decision

Inputs: {`operatorCode`, `kind=SUPPORT`, `jobTypeCode`, `networkType`, `techRegionCode`, `customerVipTier?`, `scheduledAt`}.

Output: `ContractorAssignmentDecision` { `contractorId`, `teamId`, `technicianId?`, `mode`, `considered[]` }.

v1.0 KE WIK rules:

Rule	Trigger	Decision
<code>r1-vip-internal-first</code>	<code>customerVipTier == "VIP" OR jobTypeCode == "VIP"</code>	<code>ctr_Z071H</code> (internal in-house) TEAM mode.
<code>r2-zuku-office-internal</code>	<code>jobTypeCode IN ("Z0I", "Z0T")</code>	<code>ctr_Z071H</code> TEAM mode (Zuku Office accounts handled in-house).
<code>r3-qcs-noc-maintenance</code>	<code>jobTypeCode == "QCS"</code>	<code>ctr_NOC_MAINT</code> (internal NOC) TEAM mode.
<code>r4-hfc-rf-noc-first</code>	<code>jobTypeCode == "HS3"</code>	<code>ctr_NOC_MAINT</code> TEAM mode (RF/signal-level issues go to NOC for line investigation).
<code>r5-gpon-by-region</code>	<code>networkType == "GPON" and region matches internal coverage</code>	<code>ctr_Z071H</code> TEAM mode.
<code>r6-hfc-by-region</code>	<code>networkType == "HFC" and region matches internal coverage</code>	<code>ctr_Z071H</code> TEAM mode.
<code>r7-fallback-default</code>	Otherwise	<code>ctr_ADRIAN</code> (external) TEAM mode.

4.2 site-visit-decision

Inputs: {`operatorCode`, `jobTypeCode`, `customerVipTier?`}.

Output: `REQUIRES_VISIT` or `NO_SITE_VISIT`.

v1.0 KE WIK rules:

Rule	Trigger	Decision
<code>r1-vip-always-visit</code>	<code>customerVipTier == "VIP"</code>	<code>REQUIRES_VISIT</code> (VIPs always get a tech regardless of job type)
<code>r2-no-site-visit-codes</code>	<code>jobTypeCode IN ("HS4", "PS0", "PSI", "QCA", "COS", "RPT")</code>	<code>NO_SITE_VISIT</code>
<code>r3-default-visit</code>	Otherwise	<code>REQUIRES_VISIT</code>

4.3 resolution-gate

Inputs: {woId, jobTypeCode, finalReasonCode, latestSignalReading?, latestOpticalReading?, hasEscalationCandidateNote, hasAreaOutageRefNote}.

Output: RESOLVED | NOT_RESOLVED_ESCALATE | AREA_OUTAGE.

v1.0 KE WIK rules:

Rule	Decision
final_reason == "AREA_OUTAGE_OPEN" AND area_outage_ref note present	AREA_OUTAGE
final_reason == "COULD_NOT_RESOLVE_ESCALATED" AND escalation_candidate_reason note present	NOT_RESOLVED_ESCALATE
final_reason IN ("RESOLVED_ON_SITE", "NO_FAULT_FOUND", "CABLE_REPLACED", "MODEM_REPLACED", "ONT_REPLACED", "RESOLVED_REMOTELY", "MAINTENANCE_RESOLVED")	RESOLVED
final_reason == "CUSTOMER_NOT_AVAILABLE"	RESOLVED (administratively closed; not an escalation; can be reissued by CSR as a new WO)
Default	NOT_RESOLVED_ESCALATE (defensive — unknown final reasons treated as escalation candidates for human review)

5. Pattern A escalation — narrative

The Support flow's escalation pattern is fundamentally different from Installation's. Installation uses Pattern B (same wo_id, reassigned to maintenance/construction, returns to install contractor for finalize). Support uses **Pattern A: two-WO sequence** — finalize the original Support WO, raise a new QCS WO.

5.1 Pattern A rationale (workshop quote)

From the workshop transcript:

Previously the same WO was passed between contractor and maintenance — making it impossible to separately track the maintenance team's output. QCS gives maintenance their own trackable count.

So Pattern A exists for operational accountability: the contractor's WO count reflects their workload; the NOC's QCS count reflects theirs. Pattern B (Installation) doesn't have this problem because the install contractor retains ownership end-to-end; with Support, the work genuinely transfers to a different team.

5.2 Pattern A mechanics

When the contractor first-confirms the original Support WO with final_reason = COULD_NOT_RESOLVE_ESCALATED:

1. **Drools resolution-gate returns NOT_RESOLVED_ESCALATE.**
2. **Flow worker wo-mark-escalation-candidate runs.** Appends an escalation_candidate_reason note with the contractor's handoff summary ("ONT RX -32 dBm, optical fault upstream — need OLT-side check"). Sets metadata.escalationCandidatePending = true on the WO.
3. **Second-confirm proceeds.** Framework's checklist passes; WO transitions to COMPLETED.
4. wo-emit-state-event **fires** WorkOrderFinalized (framework event, status=COMPLETED).
5. wo-emit-escalation-candidate-if-needed **runs** AFTER WorkOrderFinalized. Reads wo_final_reason_catalog.triggers_escalation_event for the WO's (operator, kind, job_type, final_reason). For COULD_NOT_RESOLVE_ESCALATED this is true. Worker emits WorkOrderEscalationCandidate (framework event from DD_WO-01-FRAMEWORK §7).

The original Support WO is now closed. A separate QCS WO needs to be created — that's step 6, which has two implementations depending on whether Ticketing module exists.

5.3 QCS WO creation — v1.0 (no Ticketing module yet)

In v1.0 the future Ticketing module hasn't shipped. The QCS WO must be created **manually by the contractor** via BO UI:

1. Contractor's BO UI subscribes to WorkOrderEscalationCandidate events for their own WOs. When one arrives, UI surfaces a prompt: "Original WO finalized as escalation. Create QCS handoff to NOC?"
2. Contractor confirms via UI button. UI calls:

```
POST /api/work-orders HTTP/1.1
```

```

{
  "operatorCode":      "WIK",
  "kind":              "SUPPORT",
  "jobTypeCode":       "QCS",
  "customerId":         "<copied from original>",
  "homepassId":         "<copied from original>",
  "originatingContextType": "MASTER_WO",
  "originatingContextRef": "<original wo_id>",
  "masterWoId":         "<original wo_id>",
  "linkType":           "ESCALATION",
  "idempotencyKey":      "qcs-from-<original-wo-id>",
  "metadata":           {"escalationCandidateNoteRef": "<note_id>"}
}

```

The new QCS WO starts a fresh Camunda process instance (same Support BPMN). The auto-assignment Drools picks ctr_NOC_MAINT (rule r3-qcs-noc-maintenance). NOC owns the new WO.

5.4 QCS WO creation — future (with Ticketing module)

When Ticketing module ships, the option to use automatic spawning opens. Ticketing subscribes to WorkOrderEscalationCandidate events and:

1. Reads the event payload (carries the original wo_id, customer, homepass, final reason, last findings note).
2. Decides whether to spawn QCS based on operator-configurable Drools rules in the Ticketing module.
3. If yes: Ticketing calls the same POST /api/work-orders endpoint above with
originating_context_type=TICKET (the new ticket Ticketing opened) and master_wo_id set to the original.
4. Returns success or "decided not to escalate" with a reason captured on the ticket.

Both paths coexist — contractor can still raise manual QCS via BO UI; Ticketing can auto-create. If both fire for the same source event (race), the framework's idempotency on (operator_code, originating_context_type, idempotency_key) deduplicates — Ticketing's auto-create uses one idempotency key, contractor's manual uses a different one, so they could in theory both succeed. Best-practice: Ticketing's auto-create includes the original wo_id in the idempotency key (qcs-auto-from-<original-wo-id>); contractor's manual uses the same key (qcs-from-<original-wo-id>). Same key → second call returns 409 IDEMPOTENCY_HIT. The flow doesn't enforce which side wins; it only ensures one QCS exists per source escalation.

5.5 What the QCS WO does

The QCS WO is just another Support WO with job_type=QCS. It runs through the same Support BPMN as any other Support WO. Differences from a regular Support WO:

- Assigned team is ctr_NOC_MAINT (internal NOC), set by Drools contractor-assignment-decision.
- master_wo_id is set to the original Support WO + link_type=ESCALATION.
- NOC verifies optics / signal levels remotely first. If they can resolve remotely →
final_reason=MAINTENANCE_RESOLVED, close.
- If they cannot resolve remotely → dispatch maintenance team physically. Same QCS WO, reassigned (TEAM-mode reassignment) within NOC.
- Workshop is explicit: **"Once maintenance resolves: updates and closes the QCS WO. No further WO raised back to the contractor."**

QCS does NOT trigger a Pattern A escalation back. If maintenance escalates further, it's a separate operational matter outside the WO flow (engineering ticket, area-outage declaration, customer compensation review — handled by other modules, not by spawning more QCS WOs).

5.6 Cancellation during escalation

If the contractor cancels the original Support WO before second-confirm, no escalation candidate event fires. If the original WO is cancelled AFTER second-confirm (which means it's already COMPLETED — cancellation from COMPLETED is not allowed per framework state machine), the situation doesn't arise. If a QCS WO is cancelled, the original Support WO is unaffected — they're independent.

6. Warranty linkage + RPT WO

Workshop: when a second WO is raised within the warranty window of a prior installation (6 months) or support visit (3 months), an RPT WO is **raised alongside** to track the repeat pattern. The original new Support WO progresses normally; the RPT WO exists for tracking + contractor billing implications.

6.1 Warranty detection at creation

The wo-check-warranty-linkage flow worker runs immediately after wo-validate-create for every new Support WO. It queries:

```
SELECT wo_id, kind, job_type_code, completed_at, metadata
FROM work_order
WHERE customer_id = $customerId
  AND operator_code = $operatorCode
  AND status = 'COMPLETED'
  AND completed_at >= NOW() - INTERVAL '180 days'
ORDER BY completed_at DESC
LIMIT 1
```

Then checks whether the prior WO's completion date plus its warranty_days is still in the future:

- Prior Installation WO with warranty_days=180 completed 90 days ago → warranty window is 90 days remaining → still under warranty → spawn RPT.
- Prior Support WO with warranty_days=90 completed 100 days ago → expired → no RPT.
- No prior WO within 180 days → no RPT.

If under warranty:

1. Append warranty_linkage note to the NEW Support WO with prior WO details.
2. Create an RPT WO via POST /api/work-orders:

```
POST /api/work-orders HTTP/1.1
{
  "operatorCode":      "WIK",
  "kind":              "SUPPORT",
  "jobTypeCode":       "RPT",
  "customerId":         "<same as new WO>",
  "homepassId":        "<same as new WO>",
  "originatingContextType": "MASTER_WO",
  "originatingContextRef": "<prior wo_id under warranty>",
  "masterWoId":         "<prior wo_id under warranty>",
  "linkType":           "REPEAT_WITHIN_WARRANTY",
  "idempotencyKey":     "rpt-<new-support-wo-id>",
  "metadata":           {
    "linkedSupportWoId": "<new support wo_id>",
    "priorJobTypeCode":  "<prior>",
    "priorWarrantyDays": 180
  }
}
```

The RPT WO is created with master_wo_id pointing to the **prior** under-warranty WO (not to the new Support WO). The new Support WO is referenced in the RPT's metadata. linkedSupportWoId for back-reference.

6.2 RPT WO lifecycle

The RPT WO is mostly an audit-and-billing marker. It runs through the Support BPMN but requires_site_visit=false (RPT itself doesn't generate field work — the linked Support WO is what generates the actual work). Lifecycle:

1. Created with status=PENDING, kind=SUPPORT, job_type=RPT.
2. Skips the on-site flow (no_site_visit branch).
3. Desk-resolution path captures warranty_linkage note (auto-populated by wo-check-warranty-linkage).
4. Auto-finalizes with final_reason=RPT_LINKAGE_RECORDED once the linked Support WO is finalized.
5. Emits WorkOrderRPTLinkageRecorded event (\$8) for Finance.

Finance uses the RPT WO + the linked Support WO's contractor info to apply warranty billing rules: typically the contractor isn't paid for warranty visits (or is paid at reduced rate). TRC/TRD codes apply when the warranty visit involves a disconnect/reconnect for equipment swap — Finance reads those codes from the linked Support WO's equipment bindings.

6.3 What if multiple prior WOs are eligible?

The worker picks the **most recent** prior WO whose warranty window is still open. Even if the customer has 3 prior under-warranty WOs, only one RPT is raised (linked to the most recent prior). This matches workshop intent: RPT is a "this is a repeat" marker, not a comprehensive incident history (history lives in the audit log + reporting layer).

6.4 Operator override

The 90/180-day warranty windows are operator-configurable via `wo_job_type_catalog.warranty_days`. Future operators can deploy different windows by editing the catalog rows — no flow code change.

7. Flow-specific workers

Four workers specific to this flow, plus two reused from the Installation flow.

7.1 wo-check-warranty-linkage

External task topic: `wo-check-warranty-linkage`.

Runs right after `wo-validate-create` for every new Support WO. Action:

1. Query for the customer's most recent COMPLETED WO within the last 180 days (§6.1 SQL).
2. If found AND prior WO's `completed_at + warranty_days > NOW()`: append `warranty_linkage` note to the new WO; create RPT WO via internal `POST /api/work-orders` call.
3. If not found or out of warranty: no-op.
4. Complete the Camunda task.

Lock duration: 8s (the SQL query + potential RPT WO creation).

7.2 wo-mark-escalation-candidate

External task topic: `wo-mark-escalation-candidate`.

Runs after `resolution-gate` Drools returns `NOT_RESOLVED_ESCALATE`. Action:

1. Verify an `escalation_candidate_reason` note is present (contractor must have captured why they can't resolve). If missing, fail the task with an error code that the BPMN routes to a user task for the contractor to add the note.
2. Set `metadata.escalationCandidatePending = true` on the WO.
3. Complete the Camunda task.

Lock duration: 3s.

7.3 wo-emit-escalation-candidate-if-needed

External task topic: `wo-emit-escalation-candidate-if-needed`.

Runs after `wo-emit-state-event` confirms `WorkOrderFinalized` for any Support WO. Action:

1. Read `wo_final_reason_catalog.triggers_escalation_event` for the WO's (`operator`, `kind`, `job_type`, `final_reason`).
2. If true: emit `WorkOrderEscalationCandidate` (framework event from DD_WO-01-FRAMEWORK §7) with payload including:
 - `woId`, `operatorCode`, `kind`, `jobTypeCode`, `customerId`, `homepassId`
 - `finalReasonCode`, `escalationCandidateReason` (the note content)
 - `originalAssignedContractorId`, `originalAssignedTeamId`
 - `suggestedHandoffTeam = "ctr_NOC_MAINT"` (per Pattern A rules)
3. If false: no-op.
4. Complete the Camunda task.

Lock duration: 3s.

7.4 wo-emit-support-completed

External task topic: wo-emit-support-completed.

Runs at the end of the Support BPMN, after warranty tagging. Action:

1. Read the WO's wo_equipment_ref_stub rows (if any — Support sometimes replaces hardware, e.g., modem swap, cable replacement).
2. Read the WO's warranty_linkage note (if present from §7.1).
3. Read the WO's area_outage_ref note (if present).
4. Emit WorkOrderSupportCompleted rich event (§8) with all relevant payload fields.
5. Complete the Camunda task.

Lock duration: 5s.

7.5 Reused workers from the Installation flow

The Support flow reuses two workers:

- wo-capture-equipment-bindings-at-finalize — same as DD_WO-01-FLOW-INSTALLATION §7.2. For Support WOs where hardware was replaced (e.g., faulty modem swapped → new modem installed), the binding is captured before final-confirm. Required bindings per Job Type for Support are looser than for Installation (most Support WOs replace nothing; only modem/cable swaps trigger binding capture). The flow's per-Job-Type rules: HS1/HS2/GP3/GP4 may have an INSTALLED+REPLACED pair if modem swapped; otherwise no binding required.
- wo-tag-warranty-window — same as DD_WO-01-FLOW-INSTALLATION §7.3. Support WOs get 90-day warranty (per wo_job_type_catalog.warranty_days = 90 for most Support codes). QCS, RPT, PSO, PSI, COS get 0 days (not warranty-eligible).

8. Flow-specific events

This flow extends the framework's event catalog with three new events. The framework's WorkOrderEscalationCandidate event is the central one for Pattern A and is fired by this flow (§7.3).

8.1 Event table

Event type	Topic	Emitted when	Key consumers
WorkOrderEscalationCandidate (framework event, fired by this flow)	sophix.work-orders.escalation-candidate	Original Support WO finalizes with final_reason.triggers_escalation_event=true	Contractor BO UI (prompts manual QCS creation in v1.0), future Ticketing module (auto-creates QCS), audit
WorkOrderRPTLinkageRecorded	sophix.work-orders.rpt-linkage-recorded	RPT WO finalizes	Finance (warranty billing decision), audit
WorkOrderSupportCompleted	sophix.work-orders.support-completed	Any Support WO (including QCS, RPT) finalizes	Finance (contractor invoicing with TRC/TRD aware billing), retention (filters), audit

8.2 Sample event — WorkOrderEscalationCandidate

```
{
  "eventType": "WorkOrderEscalationCandidate",
  "aggregateId": "wo_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "timestamp": "2026-05-18T16:30:00.512Z",
  "payload": {
    "woId": "wo_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "operatorCode": "WIK",
    "kind": "SUPPORT",
    "jobTypeCode": "GP3",
    "networkType": "GPON",
    "customerId": "cust_01HX...",
    "homepassId": "hp_01HX...",
    "finalReasonCode": "COULD_NOT_RESOLVE_ESCALATED",
    "escalationCandidateReason": {
      "whyNotResolved": "ONT RX power -32 dBm, optical loss upstream of customer premise",
      "handoffSummary": "Need OLT-side optical check + maintenance dispatch if line fault confirmed"
    },
    "originalAssignedContractorId": "ctr_ADRIAN",
    "originalAssignedTeamId": "team_adrian-eastlands_install",
    "originalCompletedAt": "2026-05-18T16:25:00.000Z",
  }
}
```

```

    "suggestedHandoffTeam":      "ctr_NOC_MAINT"
  }
}

```

8.3 Sample event — WorkOrderSupportCompleted

```

{
  "eventType":      "WorkOrderSupportCompleted",
  "aggregateId":    "wo_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "timestamp":      "2026-05-18T16:30:30.000Z",
  "payload": {
    "woId":          "wo_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "operatorCode":   "WIK",
    "kind":           "SUPPORT",
    "jobTypeCode":    "GP3",
    "networkType":    "GPON",
    "customerId":     "cust_01HX...",
    "homepassId":     "hp_01HX...",
    "completedAt":    "2026-05-18T16:25:00.000Z",
    "finalReasonCode": "COULD_NOT_RESOLVE_ESCALATED",
    "wasEscalated":   true,
    "wasAreaOutage":  false,
    "warrantyExpiresAt": "2026-08-16T16:25:00.000Z",
    "warrantyDays":    90,
    "linkedInstallWoId": "wo_01HZ_INSTALL_PRIOR",
    "linkedInstallWoUnderWarranty": true,
    "linkedRptWoId":   "wo_01HZ_RPT_LINKAGE",
    "finalAssignedContractorId": "ctr_ADRIAN",
    "finalAssignedTeamId":      "team_adrian_eastlands_install",
    "finalAssignedTechnicianId": null,
    "claimedByDispatcher":      "dispatcher_user_adrian_01",
    "equipmentBindings": []
  }
}

```

8.4 Sample event — WorkOrderRPTLinkageRecorded

```

{
  "eventType":      "WorkOrderRPTLinkageRecorded",
  "aggregateId":    "wo_01HZ_RPT_LINKAGE",
  "timestamp":      "2026-05-18T16:30:35.000Z",
  "payload": {
    "rptWoId":       "wo_01HZ_RPT_LINKAGE",
    "linkedSupportWoId": "wo_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "priorWoId":      "wo_01HZ_INSTALL_PRIOR",
    "priorJobTypeCode": "GIN",
    "priorCompletedAt": "2026-03-15T14:00:00.000Z",
    "priorWarrantyDays": 180,
    "priorWarrantyExpiresAt": "2026-09-11T14:00:00.000Z",
    "linkType":        "REPEAT_WITHIN_WARRANTY",
    "operatorCode":     "WIK",
    "customerId":        "cust_01HX..."
  }
}

```

9. REST API additions

The Support flow uses the framework's REST surface (DD_WO-01-FRAMEWORK §8) plus the Installation flow's /equipment-bindings endpoint (reused). No additional endpoints are needed in v1.0; QCS WO creation reuses the standard POST /api/work-orders endpoint with master_wo_id + link_type=ESCALATION payload fields, and RPT WO creation reuses the same endpoint with link_type=REPEAT_WITHIN_WARRANTY.

10. Workshop sources

The Wananchi KE Support workflow (workshop 2026-04-29 + Bildad Gitonga transcript 2026-04-30) requires:

- A Support WO per customer-reported issue, kind=SUPPORT, with Job Types spanning service calls (GP3, GP4, HS1, HS2, HS3, HS4, DLC2, SRV), proactive (PSO, PSI), quality control (QCA, QCS), and operational codes (COS, WCS, VIP, ZOI, ZOT, RPT, TRC, TRD, LNM).
- A configurable "Initial Reason" picklist per Job Type, linking the WO back to its originating ticket / service request.
- No-site-visit option for codes that resolve at the desk (workshop names "No Site Visit" checkbox).

- The standard finalization checklist (findings, solution, signal/optical readings where applicable, final reason, photos), enforced via `wo_finalization_requirements`.
- 2-step finalization confirm.
- Pattern A escalation: when the contractor cannot resolve, they finalize the original Support WO and a separate QCS WO is raised to NOC. Two WOs, not one. QCS gives NOC their own trackable count, separate from contractor's count.
- Area-outage exception: the contractor can finalize cleanly without resolving the customer if the issue is part of a known area outage. No QCS raised in this case.
- 3-month support warranty (vs 6-month install warranty). Within warranty, a new Support WO triggers creation of an RPT linkage WO alongside.
- TRC/TRD contractor billing codes for warranty visits.
- Final-reason picklist per workshop: `RESOLVED_ON_SITE`, `NO_FAULT_FOUND`, `CABLE_REPLACED`, `MODEM_REPLACED`, `ONT_REPLACED`, `CUSTOMER_NOT_AVAILABLE`, `AREA_OUTAGE_OPEN`, `COULD_NOT_RESOLVE_ESCALATED`, `RESOLVED_REMOTELY`.

11. Cross-DD coordination

11.1 Module owner

This satellite is owned by the Fulfillment & Operations team alongside the WO module parent. Depends on DD_WO-01-FRAMEWORK-v1.0 for all primitives and reuses workers from DD_WO-01-FLOW-INSTALLATION-v1.0 (equipment binding capture, warranty tagging).

11.2 Impacted modules

Module	Direction	Contract
WO-01-FRAMEWORK	this flow ← framework	Consumes all framework primitives.
WO-01-FLOW-INSTALLATION	this flow ← Installation flow	Reuses equipment-binding capture worker, warranty tagging worker, equipment-bindings REST endpoint. Reads installation WOs' <code>metadata.warrantyExpiresAt</code> for RPT linkage decisions (§6).
Future Ticketing module	this flow ↔ Ticketing	Consumes <code>WorkOrderEscalationCandidate</code> events to optionally auto-create QCS handoff WOs. Creates regular Support WOs from customer tickets with <code>originating_context_type=TICKET</code> . v1.0 ships without Ticketing — Support WOs created by CSR via BO UI.
Contractor BO UI	BO UI → this flow	Manual QCS creation in v1.0 (until Ticketing ships). Subscribes to <code>WorkOrderEscalationCandidate</code> for their own WOs. Standard claim/start/finalize for site-visit WOs.
NOC Console (BO UI)	BO UI → this flow	NOC team owns QCS WOs. Same dispatcher actions as any other Support WO.
ILM-CFG-01 (Customer Service — future)	this flow → ILM-CFG-01	Reads customer context at WO creation. v1.0 uses <code>customer_stub</code> .
RLM-CFG-01 (HomePass)	this flow ← RLM	Reads <code>homepass.tech_region_code</code> for assignment Drools input.
Future Inventory	this flow → Inventory	<code>WorkOrderEquipmentBindingsRecorded</code> events (reused from Installation flow's primitive) for replaced/installed equipment during support visits.
Finance	this flow → Finance	<code>WorkOrderSupportCompleted</code> consumed for contractor invoicing. Finance interprets TRC/TRD codes from equipment bindings for warranty billing. <code>WorkOrderRPTLinkageRecorded</code> consumed for warranty-aware billing decisions.
Future SLA engine (Phase 2)	this flow → SLA	Consumes all WO state events for SLA computation.

11.3 Boundary clarifications

- ■ Support BPMN (single-phase with no-site-visit branch), Drools KJAR (assignment + site-visit + resolution gates), operator-config seeds, Pattern A escalation flow, warranty + RPT linkage, area-outage exit.
- ■ Ticket lifecycle and customer-facing channels — owned by future Ticketing module.
- ■ Outage management — owned by NOC's separate outage process (not in v1.0 SOPHIX).
- ■ Contractor invoicing rules (TRC/TRD billing logic) — owned by Finance.
- ■ Customer billing for support visits — out of scope; support is contractor-billed where applicable, customer-free.

11.4 Dependencies (build order)

1. **DD_WO-01-FRAMEWORK-v1.0** must be shipped first.
2. **DD_WO-01-FLOW-INSTALLATION-v1.0** should be shipped before — Support reuses several workers and the warranty-window mechanism.
3. **RLM-CFG-01 (HomePass)** for tech-region lookup.
4. **BO UI** for CSR manual WO creation and contractor QCS manual creation.
5. **Future Ticketing module** is NOT a v1.0 dependency — Support flow works fully without it, with manual CSR + contractor BO UI flows filling the gap.

11.5 Migration approach

1. **DDL** — framework's tables are the only schema; this flow adds no new tables. Seed `wo_flow_config`, `wo_job_type_catalog` (23 new rows for Support codes), `wo_note_kind_registry` (5 new rows), `wo_finalization_requirements`, `wo_final_reason_catalog` rows per §3.
2. **BPMN deployment** — deploy `WorkOrder_Support_WIK.bpmn` to Camunda.
3. **Drools KJAR deployment** — deploy `wo-support-wik-v1.0.kjar`.
4. **Cutover** — CSR Console exposes "Raise Support WO" workflow; contractor BO UIs surface Support WOs alongside Installation and Shifting.
5. **Future operator onboarding** — add operator-suffixed BPMN + KJAR + config rows; no code change.

12. Runtime walkthrough — narrative

Two scenarios: a happy-path GP3 (GPON Internet Service Call) that resolves on site, and a Pattern A escalation where the contractor can't resolve and QCS is raised to NOC.

12.1 Happy path — GP3 Internet Service Call

Scenario. Customer Aïssatou Diop reports slow internet on 2026-05-18 morning. She's a GPON customer in Karen, originally installed 90 days ago. CSR creates a Support WO via BO UI.

Step 1 — CSR creates the WO. UI calls:

```
POST /api/work-orders HTTP/1.1
{
  "operatorCode":      "WIK",
  "kind":              "SUPPORT",
  "jobTypeCode":       "GP3",
  "customerId":         "cust_01HY...",
  "homepassId":        "hp_01HX_KAR",
  "originatingContextType": "BACKOFFICE",
  "originatingContextRef": "csr_user_42",
  "idempotencyKey":     "support-csr_42-cust_01HY-2026-05-18-08-15",
  "metadata":           {"customerName": "Aïssatou Diop", "contactNumber": "+254712345678"}
}
```

Framework creates the WO with status=PENDING, kind=SUPPORT, jobTypeCode=GP3. Starts Camunda process. Returns `woId=wo_01HZ...SP1`. Emits `WorkOrderCreated`.

Step 2 — CSR adds the initial_reason note.

```
POST /api/work-orders/wo_01HZ...SP1/notes
{"noteKind": "initial_reason", "contentJsonb": {"reasonCode": "SPEED_BELOW_EXPECTED", "reasonDescription": "Customer reports speed issue"}}
```

Step 3 — BPMN runs wo-check-warranty-linkage. SQL query finds a prior install WO wo_01HZ_INSTALL_PRIOR (GIN, completed 2026-02-17, warranty 180 days, expires 2026-08-16). Today is 2026-05-18 — under warranty. Worker:

1. Appends warranty_linkage note to wo_01HZ...SP1 with prior WO details.
2. Creates RPT WO via POST /api/work-orders with jobTypeCode=RPT, masterWoId=wo_01HZ_INSTALL_PRIOR, linkType=REPEAT_WITHIN_WARRANTY. RPT WO id: wo_01HZ_RPT....

Step 4 — Drools site-visit-decision runs. GP3 requires site visit. Takes the full flow path.

Step 5 — Drools contractor-assignment-decision. GPON + Karen region → ctr_Z071H + team_rapid_response, TEAM mode. Status → ASSIGNED. Emits WorkOrderAssigned.

Step 6 — Tech claims, /start, on-site work. tech_01HZ_KAREN_03 claims; calls /start; status IN_PROGRESS. Captures notes + readings:

```
POST /api/work-orders/wo_01HZ...SP1/notes
{"noteKind":"findings","contentText":"On-site test shows actual line speed 48Mbps. Customer's Wi-Fi router was overheating. Cu

POST /api/work-orders/wo_01HZ...SP1/notes
{"noteKind":"optical_readings","contentJsonb":{"ontRxDbm":-22.5,"oltRxDbm":-9.5}}

POST /api/work-orders/wo_01HZ...SP1/notes
{"noteKind":"solution","contentText":"Repositioned customer equipment for ventilation. Speed test now shows 50/10 Mbps. Cu

POST /api/work-orders/wo_01HZ...SP1/attachments category=setup_photo, file=ont_repositioned.jpg
POST /api/work-orders/wo_01HZ...SP1/attachments category=rf_optical_reading, file=speed_test_after.png
```

Step 7 — Contractor first-confirms.

```
POST /api/work-orders/wo_01HZ...SP1/finalize-first-confirm
{"finalReasonCode":"RESOLVED_ON_SITE","confirmedBy":"ctr_Z071H_dispatcher_01"}
```

Status → FINALIZATION_PENDING. BPMN advances to resolution-gate.

Step 8 — Drools resolution-gate. final_reason=RESOLVED_ON_SITE → returns RESOLVED. Equipment-binding worker runs — no hardware replaced, no bindings to capture, passes.

Step 9 — Second-confirm. Checklist for GP3 verifies notes (initial_reason ✓, findings ✓, solution ✓, optical_readings ✓, final_reason_set ✓) and attachments (setup_photo ✓, rf_optical_reading ✓). Pass. Status → COMPLETED.

Step 10 — wo-emit-escalation-candidate-if-needed runs.

RESOLVED_ON_SITE.triggers_escalation_event = false. No event. No-op.

Step 11 — wo-tag-warranty-window runs. Sets warrantyExpiresAt = completed_at + 90 days = 2026-08-16.

Step 12 — wo-emit-support-completed runs. Emits WorkOrderSupportCompleted (§8.3 sample) with wasEscalated=false, linkedInstallWoId=wo_01HZ_INSTALL_PRIOR, linkedRptWoId=wo_01HZ_RPT....

Step 13 — RPT WO finalization. The RPT WO (separate process instance) is meanwhile in its no-site-visit branch. The wo-check-warranty-linkage worker's RPT-creation step already populated its warranty_linkage note. The RPT WO finalizes when the linked Support WO finalizes (orchestration handled by the RPT WO's BPMN watching for the linked WO's completion event). RPT finalizes with final_reason=RPT_LINKAGE_RECORDED, emits WorkOrderRPTLinkageRecorded (§8.4 sample). Finance consumes both events; applies warranty billing logic (this contractor's visit is on a customer still under install warranty, so the visit may be billed at warranty-rate or zero depending on Finance's rules — out of WO module's scope).

Total elapsed: ~5 hours (CSR creates 08:15, contractor finalizes 13:25).

12.2 Pattern A escalation — GP3 with optical fault

Scenario. Customer Joseph Mwangi (GPON Eastlands, served by external contractor ctr_ADRIAN) reports complete service loss. CSR creates GP3 WO.

Step 1-2 — Creation and initial_reason. Same shape as §12.1.

Step 3 — Warranty check finds no eligible prior WO (customer's last install was 18 months ago, out of warranty). No RPT created.

Step 4 — Drools site-visit-decision: REQUIRES_VISIT. Drools assignment: external contractor ctr_ADRIAN + team_adrian_eastlands_install. TEAM mode.

Step 5 — External dispatcher acks the WO via /claim with claimedByExternalDispatcher=true. ADRIAN's dispatcher sends a freelancer.

Step 6 — Field tech on-site. Captures findings:

```
POST .../notes noteKind=findings, contentText="ONT shows LOS red light. Optical RX power -32 dBm, well below acceptable t
POST .../notes noteKind=optical_readings, contentJsonb={"ontRxDbm":-32.0,"oltRxDbm":-9.5,"distanceM":1450}
POST .../notes noteKind=findings, contentText="Customer's installation is fine on the customer-premise side. The optical
```

Step 7 — Contractor captures the escalation reason.

```
POST .../notes
{
  "noteKind":"escalation_candidate_reason",
  "contentJsonb":{
    "whyNotResolved":"Optical RX -32 dBm. Loss is upstream of customer ONT. Contractor cannot fix upstream optical without
    "handoffSummary":"Need NOC to verify OLT port 0/3/2/4 optics and dispatch maintenance if needed."
  },
  "appendedBy":"dispatcher_user_adrian_01","appendedByRole":"CONTRACTOR_DISPATCHER"
}
```

Step 8 — Contractor first-confirms with escalation final reason.

```
POST /api/work-orders/wo_01HZ...SP2/finalize-first-confirm
{"finalReasonCode":"COULD_NOT_RESOLVE_ESCALATED","confirmedBy":"dispatcher_user_adrian_01"}
```

Step 9 — Drools resolution-gate. `final_reason=COULD_NOT_RESOLVE_ESCALATED + escalation_candidate_reason` note present → returns `NOT_RESOLVED_ESCALATE`. BPMN takes the escalation branch.

Step 10 — wo-mark-escalation-candidate runs. Verifies escalation note present ✓. Sets `metadata.escalationCandidatePending=true`.

Step 11 — Second-confirm. Checklist passes (initial_reason, findings, solution, optical_readings, final_reason_set, escalation_candidate_reason present). Status → `COMPLETED`.

Step 12 — wo-emit-state-event fires `WorkOrderFinalized`.

Step 13 — wo-emit-escalation-candidate-if-needed runs. Reads catalog: `COULD_NOT_RESOLVE_ESCALATED.triggers_escalation_event = true`. Emits `WorkOrderEscalationCandidate` (§8.2 sample).

Step 14 — wo-emit-support-completed runs. Emits `WorkOrderSupportCompleted` with `wasEscalated=true`.

Step 15 — Contractor BO UI receives the escalation candidate event. Surfaces prompt to ADRIAN's dispatcher: "Original WO finalized as escalation. Create QCS handoff to NOC?" Dispatcher clicks yes.

Step 16 — BO UI calls POST /api/work-orders to create the QCS WO.

```
POST /api/work-orders HTTP/1.1
{
  "operatorCode": "WIK",
  "kind": "SUPPORT",
  "jobTypeCode": "QCS",
  "customerId": "cust_01HY...",
  "homepassId": "hp_01HX_EASTLANDS",
  "originatingContextType": "MASTER_WO",
  "originatingContextRef": "wo_01HZ...SP2",
  "masterWoId": "wo_01HZ...SP2",
  "linkType": "ESCALATION",
  "idempotencyKey": "qcs-from-wo_01HZ...SP2",
  "metadata": {"escalationCandidateReason":"Optical RX -32 dBm upstream"}
}
```

Returns `woId=wo_01HZ...QCS1`. Status `PENDING`. New Camunda process instance. Drools assignment for QCS fires rule `r3-qcs-noc-maintenance` → `ctr_NOC_MAINT` + `team_noc_maintenance`, `TEAM` mode.

Step 17 — NOC dispatcher acks via /claim. Status assigned, then NOC starts the WO (status `IN_PROGRESS`).

Step 18 — NOC investigates remotely. Verifies OLT port 0/3/2/4 optics — confirms upstream fault. Decides: needs maintenance physical visit. NOC dispatcher reassigns the QCS WO within NOC:

```
POST /api/work-orders/wo_01HZ...QCS1/reassign
{"contractorId":"ctr_NOC_MAINT","teamId":"team_maintenance_field","technicianId":null,"assignmentReason":"manual_team_assi
```

(Same `wo_id`; no Pattern A spawn from QCS — Pattern A only happens once.)

Step 19 — Maintenance tech fixes the optical fault. Captures findings + readings showing the new optical levels are acceptable (-22.5 dBm). Adds notes documenting the fix (splice repair at PBO-DKR-MAR-127, etc.).

Step 20 — Maintenance first-confirms with success.

```
POST /api/work-orders/wo_01HZ...QCS1/finalize-first-confirm
{"finalReasonCode":"MAINTENANCE_RESOLVED","confirmedBy":"maintenance_dispatcher_01"}
```

Step 21 — Drools resolution-gate for the QCS WO. MAINTENANCE_RESOLVED → RESOLVED. No further escalation. Second-confirm. Status COMPLETED. QCS finalized.

Step 22 — wo-emit-escalation-candidate-if-needed for QCS WO.

MAINTENANCE_RESOLVED.triggers_escalation_event=false. No event. Workshop quote satisfied: "No further WO raised back to the contractor."

Step 23 — wo-emit-support-completed for QCS WO. Finance now has both events: the original Support WO (ADRIAN's contractor visit, eligible for invoicing per their fee schedule) and the QCS WO (NOC + maintenance work, tracked separately for internal team productivity).

Total elapsed: ~20 hours across both WOs (original Support 11:00-13:30, QCS created 13:30, NOC investigation overnight, maintenance fix next morning 09:00-10:30). The customer was without service throughout — a metric the future SLA engine (Phase 2) will measure as "Support → resolved time" including QCS chain.

wo_assignment_history on the original Support WO shows one assignment (ADRIAN). On the QCS WO, three assignments (NOC dispatcher → NOC field → maintenance team). Two separate wo_ids, two separate audit trails, two separate contractor billings.

End of DD_WO-01-FLOW-SUPPORT-v1.0.